# Chapter 3

# Machine Learning

As we know there are different branches of artificial Intelligence and machine Learning of one of them which has become a most advance and widely used branch of artificial Intelligence in every field of this generation whether it may be either electronics, civil, mechanics, Bioengineering, medical or agriculture etc., In this chapter we will study about machine learning and its types with the most common algorithms of machine learning which are being mostly used in different scenarios and their implementation using python language has been discussed in the implementation part of this book. Here we will also see where machine learning can be used.

## I. Introduction

Machine learning is a subset of Artificial Intelligence and one of the most important part of artificial Intelligence. It is a field of Computer science which is totally differs from traditional computational approaches. In traditional approaches, a set of instruction is explicitly provided to the algorithm to solve any problem. Whereas in Machine Learning is a domain of an AI which uses a statistical analysis and can enables a system to always learn from its data without any external inputs in order to give an essential output. Due to this machine learning makes computers in advance to make model which provide decision based on input data.

Machine Learning involves a lot of algorithms which usually based on their requirements and functionalities to use where and when. To make a machine learning model we always require the data on which model can be made. Machine learning uses the algorithm which continuously learns from the data available as training data. A machine learning model is generated after the training of algorithm from the data. After the training completes, when trained model is provided with other real-life data then you can get an essential outcome. For example, in figure 3.1 Machine learning Model we use a predictive algorithm and make a model by providing it with some data, and later on we can receive prediction's based on real data as input.

In any field, a user of technology has always been benefited from Machine learning. In artificial intelligence as Machine learning has been always been the continuously developing field and addition to this there always been consideration which has to be kept in mind while learning machine learning algorithms and their methodologies or their analysis into different field.

**ITERATIVE LEARNING FROM DATA:** As machine learning models always get trained from different datasets before deployed for use. Some of the model which are online and can get adapted to new data continuously. On other side, some machine learning model are offline which get trained and once deployed somewhere cannot be change and they always give the output based on those. The online models in machine learning performs iterative learning as they get new data continuously in which algorithms creates patterns and associates it with data elements present in the data. After a model gets trained it can be utilized for real time to learn from data and give us meaningful outcomes.

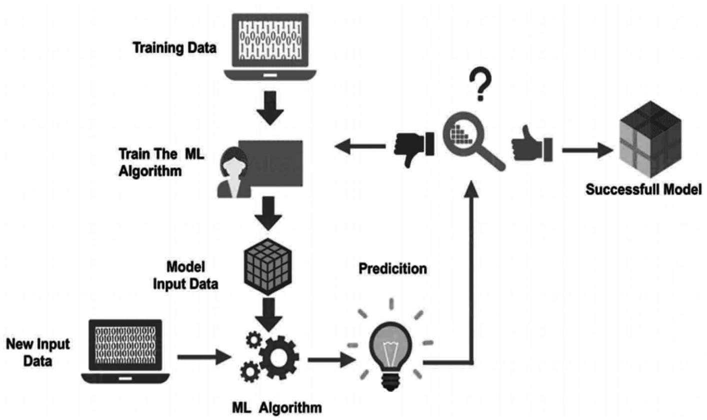


**Fig. 3.1** Machine Learning Model

## II. Approach to Machine Learning

When we talk about several type of problems to be solve by machine learning which involves different type of algorithm. In the same way we have different techniques in machine learning which can be utilized in different scenarios. For different problem statements we use different machine learning techniques which can be based on type and volume of data. Machine learning has different categories which are discussed here:

1. **Supervised Learning**

   It is the most commonly and widely used Machine Learning, Model building is always depending on data available and type of output we require. As in supervised learning, the models are given with the already classified and labeled data for training purpose or to learn and later on by computing model with other actual data, the results are compared with the assumed or expected trained one's. If certain mismatch or error is found model is again trained with more data. As supervised learning always used to predict based on the patterns found into the data or on unlabeled data given to the model.

   In other words, supervised learning can said to be a guided learning or teacher's learning where we have an already labelled dataset which acts as teacher to the Machine learning model to learn. If the model is trained, it will begin to predict or determine if new data are given. The details can be understood:

   Suppose we have input variables, $x$ and output variable, $y$, to learn how the mapping function can be learned from input to output,

   such as: $Y = f(x)$

Now, the primary objective is to approximate the mapping function in a way that the output variable (Y) can be predicted for such data if we have new input (x).

Two types of problems can be classified into primarily Supervised learning:

***Classification:*** A classification query is named when we have categorized performance such as "black," "teaching," "not teaching," etc.

***Regression:*** A regression problem is named if the real output value, like "width," "kilogram," etc., is available.

Examples of supervised machine learning algorithms are the decision tree, random forest, knn, logistic regression.

A common application of supervised learning is to use historical data to statistically predict future events. You can either forecast upcoming movements using historical stock market knowledge or use spam mails to filter. Tagged plants or crops images in agriculture can be used as input data in supervised training for classifying untagged images of plants and crops.

2. **Unsupervised Learning:**

Unsupervised learning is best if the question involves a large amount of unlabeled data. Knowing the meaning behind these data includes algorithms that can be understood by classifying the data on the basis of the patterns or clusters found. The supervised learning thus carries out an iterative data processing process without human intervention.

The Algorithms in Unsupervised learning segment data into outstanding groups (clusters) or feature groups. Unlabeled data determines the values of the parameter and the data classification. This method basically applies labels to the information so that it is becomes supervised and model gets trained. Unsupervised learning will decide whether a large amount of data is available. The developer does not know the context of the data to analyze in this case, so at this point, the labeling is not possible. That's Why Unsupervised learning can therefore be used as the first step before data are moved to a regulated learning process. The details can be understood:

Suppose we have an input variable x, and no output variables are available as in supervised learning algorithms. Simply put, we may conclude that there is no right answer and no guidance instructor in unsupervised learning. Algorithms allow users to identify fascinating data trends.

The following two forms of problem can be divided into unsupervised learning problems:

***Clustering:*** We must discover the underlying groupings of the data in problems of clustering. For instance, consumer grouping by shopping behavior.

***Association:*** An association problem is named since these problems require that rules be discovered that explain many of our data for instance, to find the clients that buy both x and y.

In unsupervised learning algorithms can help understand large amounts of new, unlabeled content. These algorithms search for data patterns similar to supervised learning (see the preceding Section). The difference is that data are not already known.

*For example*, The Unsupervised machine learning algorithms are like K-means for clustering, Apriori algorithms for association.

In agriculture field, gathering vast quantities of data on a particular field will, for example, help farmers gain insight into trends and link them to crop harvesting and increase in production of it. All the data points linked to a condition like health of plants will take too long to classify. The unregulated learning strategy will also help to evaluate outcomes faster than a supervised approach to learning.

3. **Reinforcement Learning**:

Reinforcement learning is a model of behavioral learning. Feedback from the data analysis is received in order to guide the user in the best result. Increased learning differs from others because the system does not use a sample data set to train. Instead, the system learns by testing and error. A series of successful choices will therefore "reinforce" the process as the problem is best resolved.

Figure Agent→system

In the Figure which reflects the general scenario of improving learning. In comparison, the agent receives no designated data here, unlike the supervised learning scenario considered in the previous section. It gathers information instead by communicating with the world, via a series of acts. The agent receives two kinds of information in response to an event: his or her state of play in the world and the real-life scenarios, which is unique to the task and its respective purpose. The goal is for the Agent to maximize its reward and, in order to achieve that objective, to agree on the best course of action or policy.

The data he collects from the system, however, is just the immediate reward for the action he has taken. There is no input from the system on future or long-term incentives. A significant element of enhancement is to take account of late bonuses or penalties. The agent faces the dilemma between exploring unknown states and actions to gain more information about the System and the rewards and using the already accumulated information in order to maximize its reparation. This is known as the trade-off between exploration and development inherent in Reinforcement learning.

Understand that in several of the preceding chapters there are many variations between the reinforced learning scenario and the supervised learning. In comparison to supervised learning, there is no fixed distribution of the instances in reinforcement learning, which defines a distribution over observations. It is the choice of policy. In reality, minor policy changes can drastically affect rewards. In addition, the System cannot be set in general and can differ due to the chosen behavior by the agent. This may be more practical than normal supervised learning for certain learning problems.

There are two key settings: one in which the ambient model is known to the agent, which reduces its goal of optimizing the amount of money to a planning problem; and the other in which the ambient model is unknown and where the agent is faced with a learning question. In the above case, the agent must learn from the state and recompense data obtained to acquire relevant information and decide the best strategy.

*Reinforcement Learning is not specifically supervised Learning as not strictly reliant on collection of data "supervised" (or labeled) (training collection). In reality, the response of actions taken can be controlled and calculated against a concept of "reward." However, it is also not unsupervised learning because we know the desired reward in advance when we form our "learner."*

## III. Supervised Vs Unsupervised Vs Reinforcement

Finally, as all are well aware of supervised, unsupervised and Reinforcement learning, look at the gap between these learning's.

To sum up, supervised learning happens when a model learns with guidance from an defined data set. And unsupervised learning is where the computer is trained without any guidance on the basis of unlabeled data. Reinforcement learning happens when a computer or an agent communicates, carries out actions and learns using a test-and-error process.

| Criteria | Supervised Learning | Unsupervised learning | Reinforcement Learning |
|---|---|---|---|
| Definition | The Model learns using labelled or classified data | The model is trained with unclassified or unlabeled data | An agent interact with the model by taking action & learning from the errors and rewards. |
| Types of Problem | Regression & Classification | Association & Clustering | Reward based |
| Type of data | Labeled Data | Unlabeled data | No predefined data |
| Training | External Supervision | No supervision | No supervision |
| Approach | Maps the labeled or classified inputs to the known outputs | Recognizes the pattern in unclassified data and discovers the output | It follows the trail and error method |

## IV. Most Common Machine Learning Algorithms

- **Linear Regression:** This is one of the most popular statistics and machine learning algorithms.

  Linear regression is essentially a model of a linearity that assumes a linear relationship between the input variables $x$ and $y$. In other words, you can calculate the input variables $x$ from a linear combination. It can be defined by a best line to create a relationship between the variables. Linear Regression is generally of two types which are

  - ***Simple Linear Regression:*** A linear regression algorithm is considered a simple linear regression because there is just one independent variable.

- ***Multiple Linear Regression:*** If it has more than one independent variable, a linear regression algorithm is considered a multiple linear regression.

  Linear regression is used mainly for estimating true values based on continuous variable (s). For example, a full day sale of a shop can be calculated by a linear regression based on real values.

- **Logistic Regression:** It is also known as logit regression. It is a classification algorithm. Logistic regression is mainly a classification algorithm used to estimate discrete values like 0 or 1, true or false or yes or no based on a particular collection of separate variables. In theory, it predicts that its output will be between 0 and 1.
- **Decision Tree:** It is a supervised learning algorithm that is used mainly for problem classification. This is basically a graded partition based on the independent variables as a recursive partitive. Decision tree has rooted tree nodes. The Rooted Tree is a directed tree with a root node. Root has no input edges and all the other nodes have an input edge. Such nodes are referred to as leaves or nodes of judgment. Consider, for example, the following Fig 3.2 decision tree to test whether or not an individual is suitable for the job.

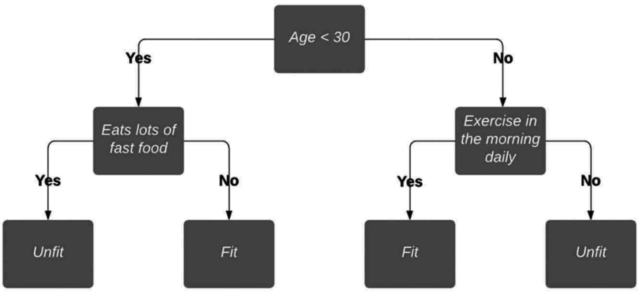


**Fig. .3.2** Decision Tree

- **Support Vector Machine:** It is used for problems of classification and regression. However, it is primarily used for problems of classification. SVM's key principle is to assign each data element in the *n*-dimensional space as the value of the individual feature. Here n are the characteristics we have. Then a simple graphic representation to grasp the SVM definition from the Fig 3.3 SVM.

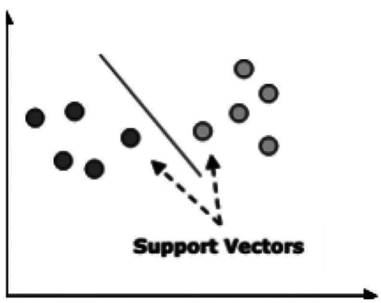


**Fig. 3.3** SVM

Within the above Fig 3.3 SVM, we have two features hence we first need to map such two variables within two-dimensional space where each point has two co-ordinates, called support vectors. The line divides the data into two separate classified categories..This line is the classifier in SVM

- **Naïve Bayes:** It's also a technique of classification. Bayes theorem is a principle for constructing classifiers behind this classification technique. The predictors are expected to be independent. In simple terms, the inclusion in a class of a particular characteristic is not connected to that of any other characteristic. The following is the Bayes theorem equation:

  $$P(A|B) = P(B|A)\ P(A)/P(B)$$

  The model of Naïve Bayes is simple to develop and especially useful for large data sets
- **K-Nearest Neighbors (KNN):** It is used to identify problems and to rectify them. It is used commonly to solve problems with classification. The key principle of this algorithm is that it stores all the existing cases and by voting plurality of its neighbors classifies new cases. The case is then assigned to the class which, according to a distance function, is the most common among his nearest K neighbors. The gap can be from Minkowski, Euclidean, and Hamming. Consider KNN as follows:
  - Computationally KNN are expensive than other algorithms used for classification problems.
  - The normalization of variables needed otherwise higher range variables can bias it.
  - At KNN, we will work on a stage such as noise reduction, pre-processing.
- **K-Means Clustering:** This is used to resolve the problem of clusters, as the name implies. This is a kind of unsupervised Learning. The principal principle of the K-Means algorithm is to assign the data set across different clusters. Follow these steps to build K-means clusters:
  - For each cluster called centroids, K-means picks up the k number of the points.
  - Every data point now forms a cluster, i.e., k clusters, with the nearest centroids
  - Now, the centers, based on the current members of each cluster, will be identified.
  - These measures need to be repeated before convergence takes place.
- **Random Forest:** This is a supervised algorithm of classification. The random forest algorithm has the advantage that it could be used for problems of classification as well as regression. This is essentially the set of decision-making trees (forests) or the decision-making trees ensemble. The basic principle of the random forest is that each trees are graded and that the forest selects the best grades. The advantages of Random Forest algorithm are as follows:
  - For classification and regression functions, the random forest classifier may be used.
  - The missing values can be managed.
  - Even if we have more trees in the forest, this will not over suit the pattern.

## V. Applications

- Retailers

  Machine Learning in agriculture is used by the seed retailers to turn data into better crop production. Through their use by pest control firms, the growing bacteria, bugs and vermin can be detected.

- Agriculture Bots

  The majority of companies are now preparing and building robots for the main agricultural mission. This involves seed processing and work more efficiently than manpower. This is the latest example of agricultural machine learning.

- Species Breeding

  Species selection is a repetitive searching method for specific genes, which decide the effectiveness of water and the use of nutrients, climate tolerance, disease resistance, nutrient content or taste. In the analysis of crop production in different climates and developing new technologies, machine-learning in particular, profound learning algorithms requires decades of field data. Based on this data, a statistical model can be developed that predicts which genes are most likely to contribute to a plant's gain.

- Species Recognition

  The conventional approach of the human person for plant classification is comparing the color and shape of the leaves, but by using Machine Leaning's statistical analysis can provide quicker and more detailed results by analyzing the morphology of the leaves and provides more details about the leaf properties.

- Yield Production

  Yield forecasting is one of the topics of steep-precision agriculture, as it defines yield mapping and prediction, crop demand matching and crop management. state-of - the-art techniques have gone far beyond simple prediction based through historical data, but have involved machine learning techniques to provide knowledge concerning crops, weather and economic circumstances and extensive multidimensional analysis to optimize the return for farmers and the public.

- Disease Detection

  Machine Learning can be used to detect Plant infections which are typically caused by plague, insects, diseases and, unless managed on time, they reduce productivity on a large scale. Concerning the area grown in acres, the cultivators are tedious to track the crops daily. By implementing Machine Learning here provides the solution to monitor the cultivated area consistently and provides automatic detection of diseases with remote sensing images.

- Crop Quality

  Data output can be adequately measured and defined to lift product price and reduce waste. Contrary to human experts, machines may use apparently irrelevant data and interfaces to discover and identify new qualities that play a role in the overall quality of the crops.

- Weed Detection

  Weeds are the major threats to crop production, apart from diseases. The main issue in the war against weeds is that they are hard to distinguish and discriminate against crops. Computer vision and ML algorithms are able to enhance weed identification and discrimination at low cost without environmental issues or side effects. The technologies are going to push robots to kill weeds and reduce the need for herbicides in future.

- Soil Management

  Soil is a highly diverse natural resource with complex processes and unclear mechanisms for agricultural specialists. Its own temperature can give insights into the impact on territorial returns of climate change. The algorithms of machine learning study the process of evaporation, soil humidity and temperature to understand ecosystem dynamics and the effects of agriculture.

- Water Management

  The hydrological, climatological and agricultural balance of water resources in agriculture has a significance. Using more efficiently irrigated systems and a forecast of a daily dewpoint temperature, the most advanced ML applications so far are connected with an estimation of daily, weekly, or monthly stomata, which helps to classify predicted weathers and to estimation the evaporation.

- Livestock Production

  Machine learning offers accurate prediction and calculation of farming parameters, in accordance with crop management, to maximize the economic efficiency of farming systems such as livestock production. In order to allow farmers to change diets and conditions, for example, weight forecasting system will be estimating future weight 150 days before the day they slaughter.

## Summary

As Machine learning has been widely used in several field and it has become a most important branch of artificial Intelligence. As we have seen there can be a lot of used of Machine Learning in the field of agriculture which can help the farmers to increase their production of crops and save them from pests and other insects while harvesting in the field. So, we can see that how machine learning can be useful in the agricultural field.